Yiming Chen

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Education

Georgia Institute of Technology — Atlanta, GA
Ph.D. in Machine Learning — 2026 - Present
- Advisor: Prof. Siddharth Karamcheti

Georgia Institute of Technology — Atlanta, GA
Master's Degree in Computer Science — 2024 - 2026
- GPA: 4.00/4.00
- Advisor: Prof. James Hays

New York University — Manhattan, NY
Bachelor's Degree in Computer Science and Mathematics — 2020 - 2024
- GPA: 3.73/4.00
- Advisor: Prof. Gizem Kayar-Ceylan

Publications

1. Liu, Y., **Chen, Y.**, & Zheng, M. (2026). Do Agentic Policies Need to Look at Every Step? Geometric Abstraction for Reasoning and Action. *CoRL 2026 Workshop on Agentic Robotics*.
2. **Chen, Y.**[*], Singh, S.[*], Chatterjee, A., Raj, A., Hays, J., Yang, Y., & Baral, C. (2025). Chimera: Compositional Image Generation using Part-based Concepting. *arXiv preprint arXiv:2510.18083*.
3. Yang, M., **Chen, Y**., Pei, H., Agarwal, S., Vasudevan, A. B., & Hays, J. (2025). Clink! Chop! Thud! - Learning Object Sounds from Real-World Interactions. *ICCV 2025*.
4. **Chen, Y**., Liu, Y., & Kayar-Ceylan, G. (2025). CSG-based ML-supported 3D translation of sketches into game assets for game designers. *The Visual Computer*, 1-13.
5. **Chen, Y**. (2020). A Device for Compressing Garbage in Dustbin. *China National Intellectual Property Administration*, Patent No. CN210823859U.

Presentations

1. **Chen, Y.**, Liu, Y., "CSG-based ML-supported 3D translation of sketches into game assets for game designers", New York University Undergraduate Research Conference, NY, May 2024

Research Experience

Egocentric-to-3D Interaction Reconstruction
Research Assistant, Advised by Prof. Yalong Yang — Mar 2025 - Present

- Implementing a 3D reconstruction pipeline using MegaSam to map egocentric interaction data into semantically segmented 3D spaces.
- Developing a novel view synthesis framework that decouples the observer from the user, allowing third parties to visualize first-person tasks from arbitrary external viewpoints.

Gaussian Visibility Field for Uncertainty Mapping
Research Assistant, Collaboration with NVIDIA — Aug 2024 - Dec 2024

- Developed novel neural field representations by applying Bayesian deep learning principles to map and quantify uncertainty in 3D scenes for autonomous systems.
- Engineered the system for real-time performance by optimizing the neural field's computational efficiency through advanced GPU parallelization and CUDA techniques.

Unity ML-Agents with Walking Robot using Curriculum Training
Research Assistant, Advised by Prof. Jeremy Curuksu — Mar 2024 - May 2024

- Investigated sample-efficient training for embodied agents by integrating curriculum learning with advanced deep RL algorithms (PPO) to accelerate policy convergence.
- Enhanced the agent's navigation capabilities by implementing a ray-based perception system for robust and efficient obstacle avoidance in complex environments.

Industry Experience

Amazon Web Services - Annapurna Labs — Cupertino, CA

Machine Learning Engineer Internship — May 2026 - Aug 2026

- Built an on-device FP8-static calibration and inference pipeline for vLLM-Neuron on AWS Trainium, supporting Llama 3.x, Qwen3-MoE (up to 235B), and Qwen3-VL.
- Scaled calibration with a (DP, TP) device mesh and shard-on-read loading, up to 7.1x faster on Llama-70B, calibrating Qwen3-235B in ∼32 minutes within 0.02 GSM8K accuracy of BF16.
- Implemented data-free block-scale MXFP8 quantization for Qwen3-VL-8B, reducing model size by 45% at matching accuracy, and landed end-to-end CI accuracy gates on TRN2.

Machine Learning Engineer Internship — May 2025 - Aug 2025

- Designed and implemented pipeline parallel processing in NxDI stack for large language models, improving distributed training efficiency by 35%.
- Implemented Sequence Pipeline Parallel to create a prompt parallel architecture for window context encoding, enhancing vLLM inference performance by 2.3x.
- Collaborated with the inference team to optimize distributed tensor and pipeline parallelism, documenting 40% latency improvements using NeuronTransferEngine.

T.C.L. Industries Holdings — Shenzhen, China

Machine Learning Engineer Internship — Jun 2024 - Aug 2024

- Led development of LSTM-based predictive models, raising accuracy from 40% to 90%.
- Developed APIs for efficient internal use of the model, reducing costs by 35%.
- Integrated the model with Microsoft 365, enabling teams to fine-tune task-specific models.

Awards

- **Dean's Undergraduate Research Fund**, New York University — 2023.09
- **Registration Support Scholarship**, New York University — 2022.09
- **Student Relief Scholarship II**, New York University — 2022.02
- **Student Relief Scholarship**, New York University — 2021.09

Teaching

Teaching Assistant — Georgia Institute of Technology

Computer Vision (CS 6476 A) - Head TA	Spring 2026
Computer Vision (CS 6476 A) - Head TA	Fall 2025
Computer Vision (CS 6476 A)	Spring 2025

Teaching Assistant — New York University

Computer Systems Organization (CSCI-UA 201)	Spring 2024
Computer Systems Organization (CSCI-UA 201)	Fall 2023
Great Ideas in Math (CORE-UA 110)	Spring 2023
Algebra, Trigonometry, and Functions (MATH-UA 9)	Fall 2022